8.3 Practice: volume and angles

← Unit overview · Answers

These sets start easy and build up. Keep the "1 dm³ = 1 litre" box from activity 1 in mind. There is no clock.

Helpful table: 1 l = 10 dl = 100 cl = 1 000 ml · 1 dm³ = 1 l · 1 cm³ = 1 ml · 1 m³ = 1 000 dm³ = 1 000 l

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Set A: Litres and friends (Warm-up)

Fill in the gap.

1. 1 l = … ml
2. 2 l = … ml
3. 500 ml = … l
4. 1 dm³ = … l
5. 1 cm³ = … ml
6. 3 dm³ = … l
7. 250 ml = … dl, and 250 ml = … cl
8. 1,5 l = … ml
9. 1 m³ = … dm³ = … l
10. 750 cm³ = … ml = … l
11. 1 l = … dl

Talk about it: Which one did you need the helpful table for? Which ones did you just know?

Set B: Volume of a box or cube (Building)

Volume (*inhoud*) = length × width × height.

1. A cube (*kubus*) with sides of 1 cm. Volume in cm³?
2. A cube with sides of 2 cm. Volume in cm³?
3. A box (*balk*) 3 cm long, 2 cm wide and 1 cm high. Volume?
4. A box 5 cm × 4 cm × 2 cm. Volume?
5. A cube with sides of 10 cm. Volume in cm³? How many litres is that?
6. A shoebox is 30 cm long, 20 cm wide and 10 cm high. Volume in cm³? How many litres?
7. An aquarium is 50 cm long, 30 cm wide and 40 cm high. How many litres of water fit in it, filled to the top?
8. A juice carton holds 1,5 litres. Its base is 10 cm by 6 cm. How tall is the carton (on the inside)?
9. A sandpit (*zandbak*) is 2 m long, 1,5 m wide and 0,2 m deep. How many m³ of sand fill it? How many litres is that?
10. How many 1 cm cubes fit in a box that is 4 cm long, 3 cm wide and 5 cm high?
11. A cube with sides of 3 cm. Volume in cm³?

Talk about it: For number 8, how did you work backwards?

Set C: Kinds of angles (Building)

Use: *scherpe hoek* (acute, less than 90°), *rechte hoek* (right, 90°), *stompe hoek* (obtuse, between 90° and 180°), *gestrekte hoek* (straight, 180°).

1. The corner of a book. What kind of angle? How many degrees?
2. An angle of 45°. What kind?
3. An angle of 120°. What kind?
4. An angle of 180°. What kind?
5. How many degrees between the hands of a clock at 3:00?
6. How many degrees between the hands at 6:00?
7. How many degrees between the hands at 1:00?
8. How many degrees between the hands at 2:00?
9. A full turn is 360°. How many degrees is a half turn? A quarter turn? Three quarters of a turn?
10. You cut a right angle exactly in half. How big is each part?
11. An angle of 160°. What kind?

Talk about it: What's your trick for remembering *scherp* and *stomp*? (Hint: think of a knife.)

Set D: Measure, draw and find the missing angle (Stretch)

You need a protractor (*gradenboog* or *geodriehoek*) and a sharp pencil. **Estimate first** for every angle: smaller or bigger than a right angle?

1. Draw a right angle (90°). Check it with the corner of a sheet of paper.
2. Draw an angle of 45°.
3. Draw an angle of 30°.
4. Draw an angle of 135°. What kind of angle is it?
5. Two angles sit next to each other on a straight line. One is 130°. How big is the other?
6. A triangle has angles of 90° and 40°. How big is the third angle?
7. A triangle has two angles of 60°. How big is the third angle?
8. A triangle has angles of 100° and 50°. How big is the third angle?
9. Three equal angles fit exactly around a point (a full turn). How big is each one?
10. A rectangle has four right angles. What do they add up to?

Talk about it: In numbers 6, 7 and 8, what number did you always start from?

Puzzle corner (optional, just for fun)

1. What is the angle between the clock hands at 3:30? (Careful: the hour hand doesn't stay on the 3.)
2. A 1 dm cube is filled with 1 cm cubes. How many little cubes is that?
3. You make a box twice as long, twice as wide and twice as high. How many times as much does it hold?